Mastering Agentic AI in Bioinformatics Research: From Ideas to Results

From Biostatistics to Agentic AI in Bioinformatics

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Objectives

1. Trace how biomedical data analysis evolved from biostatistics to agentic AI.
2. Understand what makes an AI system “agentic” in bioinformatics.
3. Recognize why statistical reasoning matters more as AI becomes autonomous.

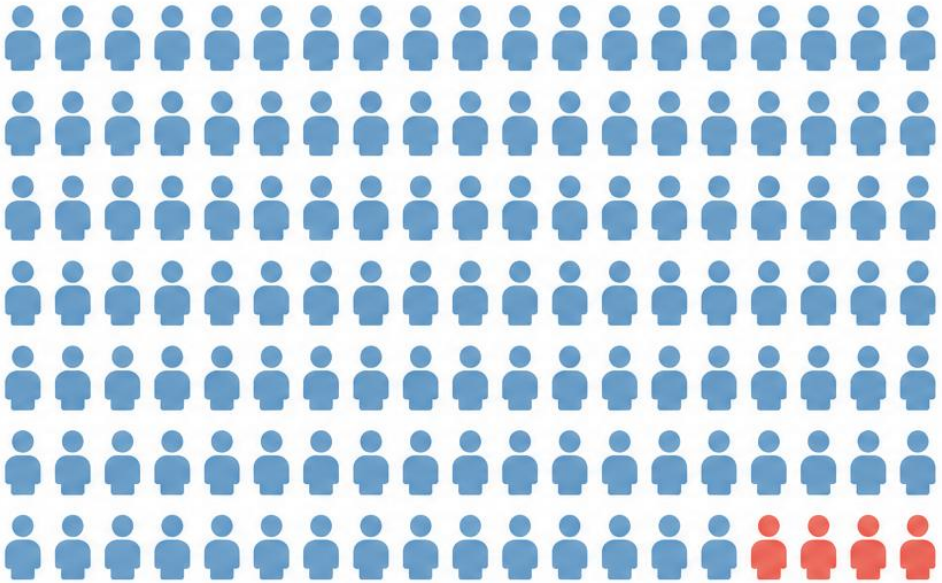
Outline

- Introduction
- Biostatistics: reasoning under uncertainty
- Bioinformatics: computing with biological data at scale
- Machine & deep learning: learning from data
- Agentic AI: acting on scientific questions
- Significant results in bioinformatics
- Why statistics matters more than ever
- Conclusion

Rare probability

5% is a rare probability
with 107 patients with pulmonary condition

Total patients: 107 (with pulmonary condition)



Not the event (95%)
= 102 patients

Event (5%)
= 5 patients

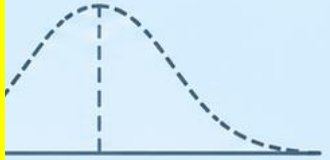
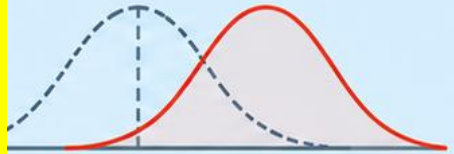
A 5% probability means
only 5 out of 107 patients
are expected to have the event
(on average).

For comparison:

5%	→	5 out of 107
10%	→	11 out of 107
20%	→	21 out of 107
50%	→	54 out of 107

So, 5% is a rare probability in a group of 107 patients.

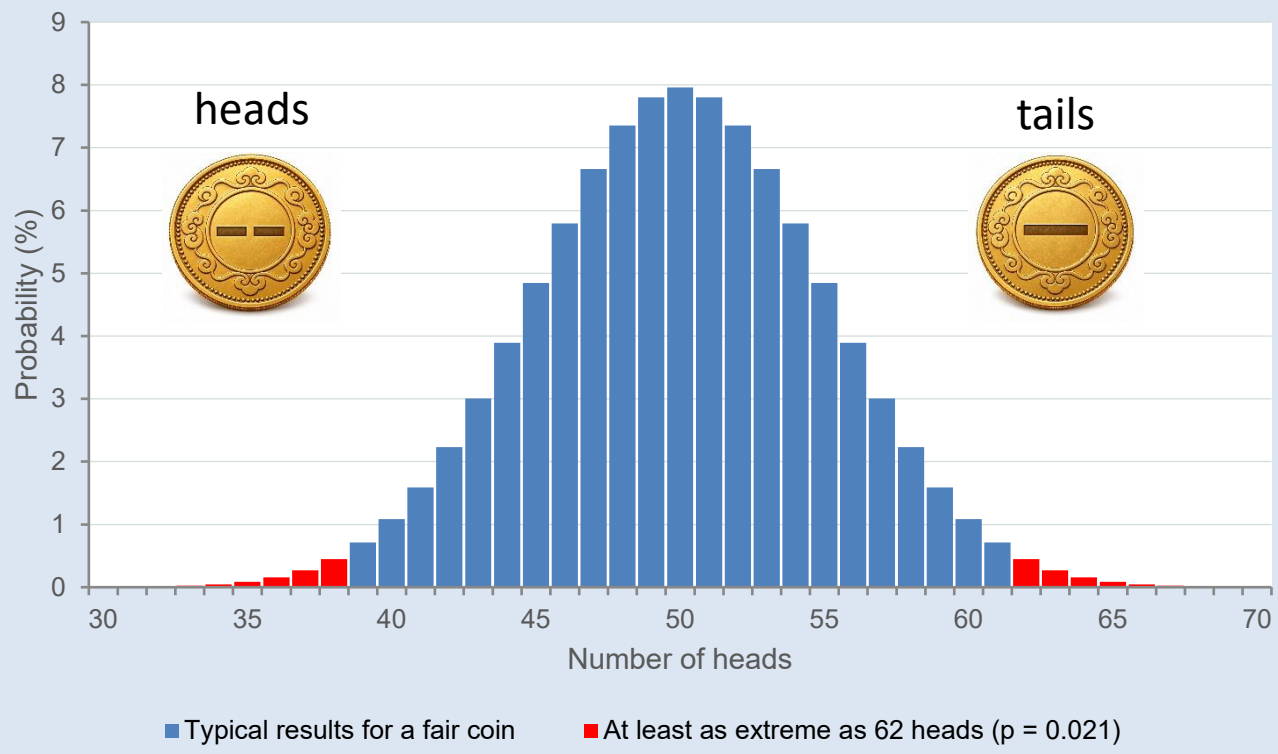
Hypothesis testing

IN COURT		IN SCIENCE
	Defendant presumed innocent	**Null hypothesis H_0** "Nothing is going on": no difference, no effect, no association. 
	Prosecution claims guilt	**Alternative hypothesis H_1** "Something is going on": the drug works, the gene matters. 
	Evidence is presented	**Data** Measurements from an experiment or a study. 
	Jury weighs the evidence	**Test statistic and p-value** How surprising are these data if H_0 were true? 
	Verdict	**Reject H_0 — or fail to reject** "Guilty" = reject H_0. "Not guilty" ≠ innocent: we simply lack proof. 

References: Fisher 1925; Neyman & Pearson 1933 (null/alternative framework).

A worked example: is this coin fair?

100 flips of a fair coin: how often does each count occur?



Binomial distribution, n = 100, p = 0.5 (own calculation).

H_0 The coin is fair (P(heads) = 0.5).

H_1 The coin is not fair.

Data 62 heads out of 100 flips.

p

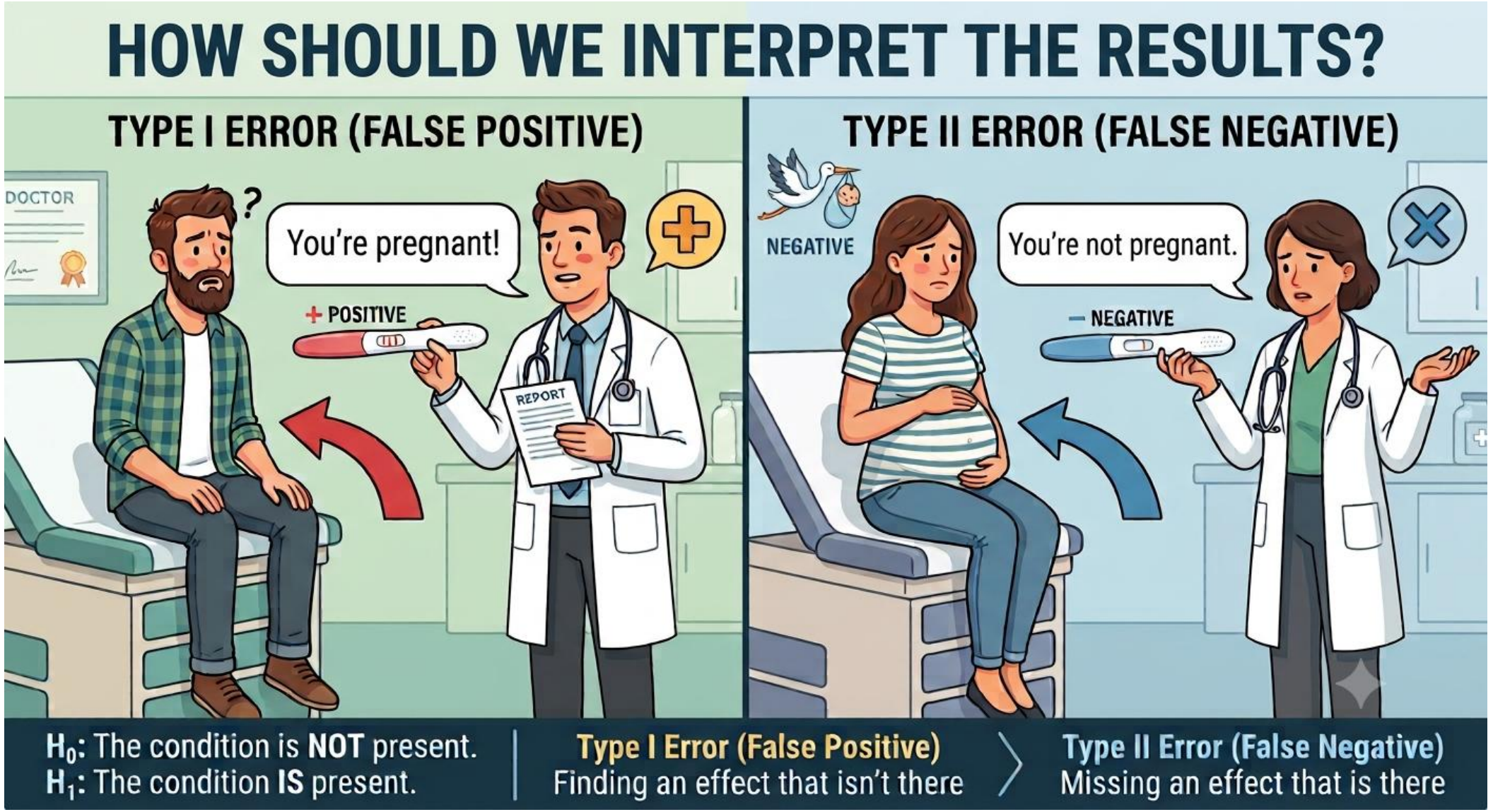
$$P(62+ \ or \ 38-)$$
$$= \binom{100}{62} 0.5^{62}(1-0.5)^{38}$$
$$= \binom{100}{62} 0.5^{100}$$
$$\approx \mathbf{0.021}$$

Decision $0.021 < 0.05 \rightarrow$ **reject H_0**

The coin is **BIASED**

References: Binomial test; Fisher 1935. Counts and p-value: own calculation.

Five steps, and two ways to be wrong (1/2)



Five steps, and two ways to be wrong (2/2)

THE FIVE STEPS

1. State the hypotheses (H_0, H_1)
2. Choose α before seeing the data
3. Collect the data
4. Compute the test statistic and p-value
5. Decide: reject H_0 or not

TWO WAYS TO BE WRONG

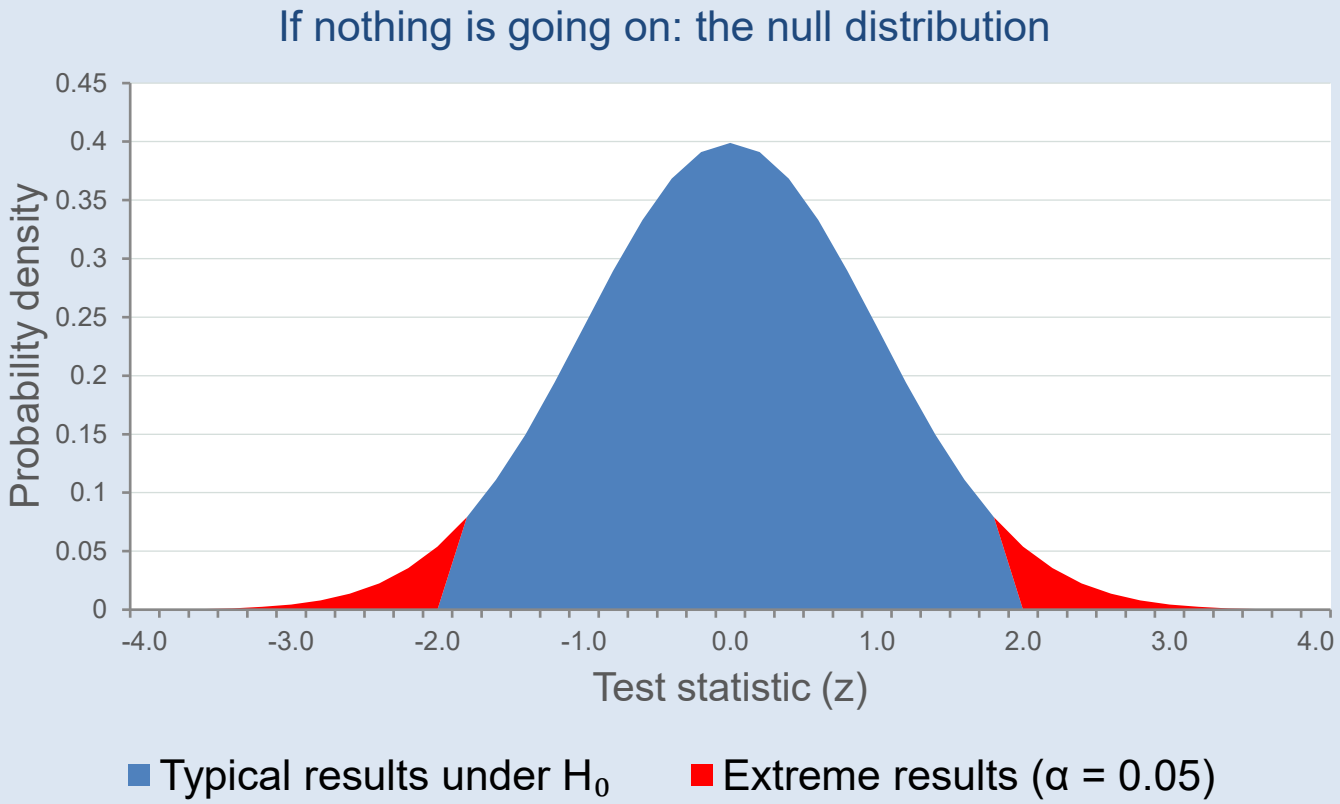
	Reality → **H_0 is true** (drug does nothing)	**H_0 is false** (drug works)
Decision: Reject H_0	**Type I error** false alarm; probability $= \alpha$	**Correct** power $= 1 - \beta$
Decision: Don't reject H_0	**Correct** probability $= 1 - \alpha$	**Type II error** missed effect; probability $= \beta$

Test 20,000 genes at $\alpha = 0.05$:

≈ 1,000 false alarms by CHANCE.

References: Neyman & Pearson 1933 (Type I/II errors, power); Benjamini & Hochberg 1995.

What does “statistically significant” mean?



Red tails = 5% of the curve: results this extreme occur in only 1 of 20 experiments when nothing is going on.

1. **Assume nothing is going on**
   The null hypothesis H_0: any difference is random variation.

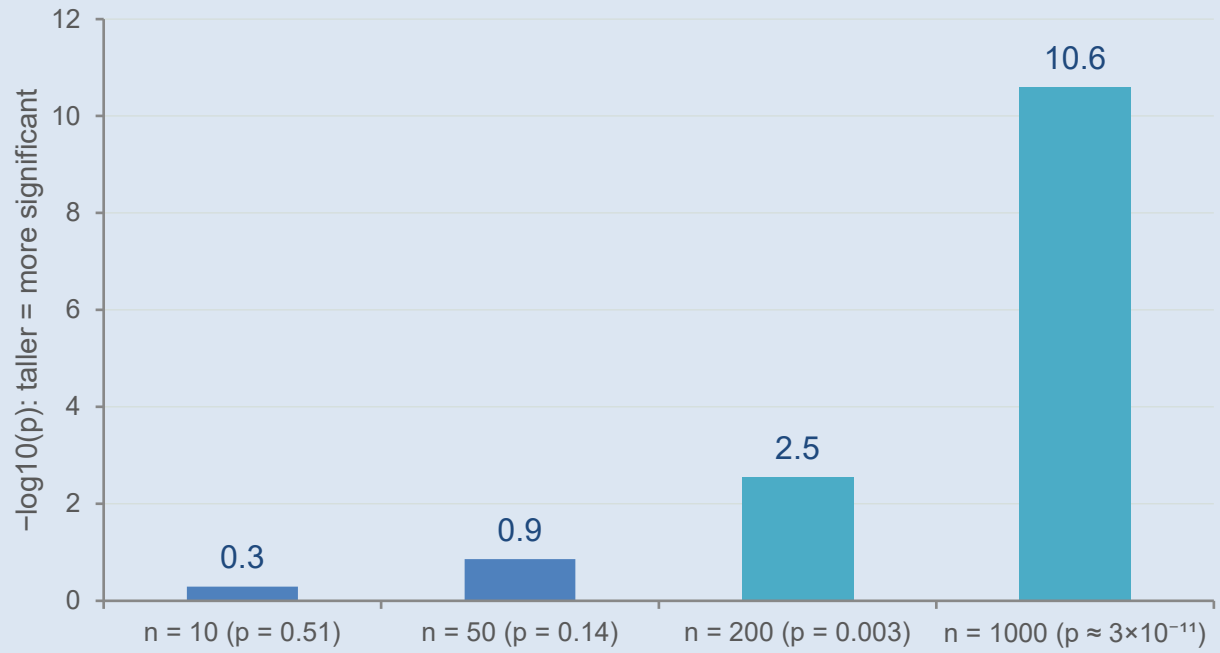
2. **Measure how surprising the data are**
   p-value = probability of a result at least this extreme if H_0 were true.

3. **Compare with a threshold set in advance**
   If $p < \alpha$ (usually 0.05), the result is “statistically significant”.

Significant = hard to explain by CHANCE.
Not “large” or “important”.

References: Fisher 1925; Neyman & Pearson 1933; Wasserstein & Lazar 2016 (ASA statement on p-values).

Reading a p-value correctly



Own calculation: two-sample t-test, standardized effect d = 0.3, n per group as shown. Significance at p < 0.05 corresponds to −log10(p) > 1.3.

References: Wasserstein & Lazar 2016; Benjamini & Hochberg 1995; Ioannidis 2005. Chart: own calculation.

A p-value is not the effect size
A tiny p can come from a trivial effect measured in a huge sample.

It is not P(H_0 is true)
It assumes H_0 and asks how surprising the data are.

p > 0.05 does not prove "no effect"
It may only mean the study was too small to detect it.

Many tests, many chances: correct for multiplicity.

Why biostatistics: a real trial (1/3)

MRC streptomycin trial, 1948

Design

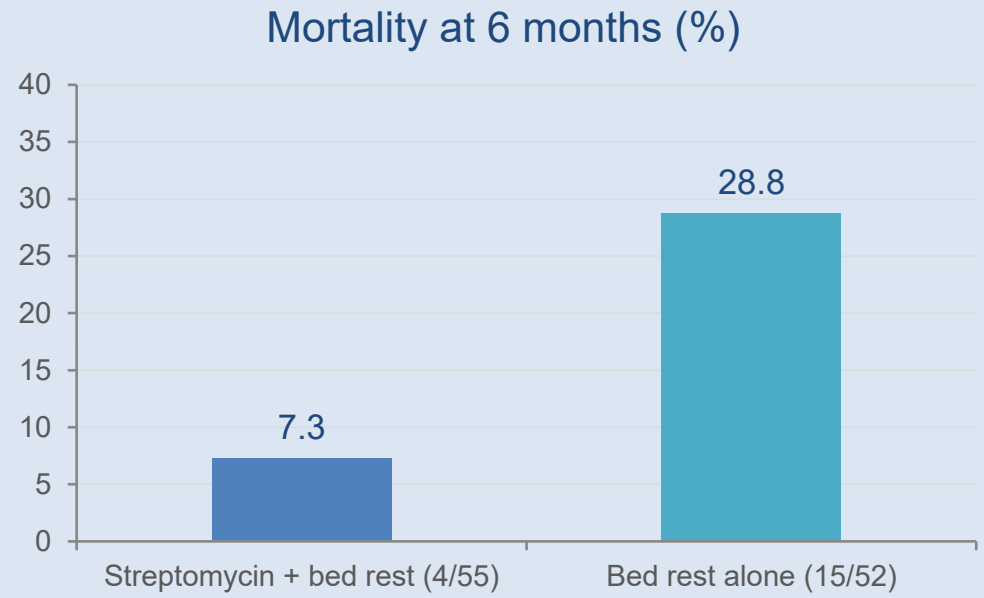
107 patients with pulmonary TB; **random allocation** by numbered sealed envelopes (55 streptomycin + bed rest, 52 bed rest alone)

Outcome

Deaths in the first 6 months: **4 of 55 vs 15 of 52**

Question

Is a difference of this size explained by chance? Biostatistics answers with an effect size, an interval, and a test.



22 pts
risk difference
95% CI 7–36

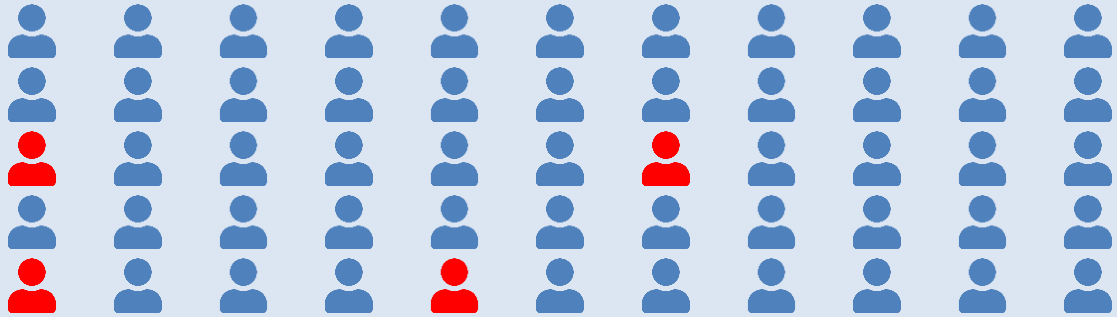
0.25
risk ratio
95% CI 0.09–0.71

Counts from the published trial as summarized by the James Lind Library; effect size, CI and p-value are my own calculation (Fisher exact test; Wald CI).

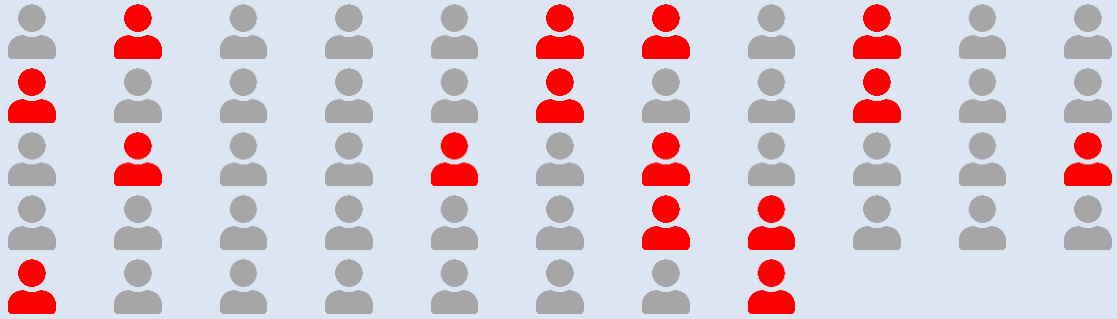
References: Medical Research Council 1948 (BMJ 2:769–82); counts via James Lind Library (jameslindlibrary.org/medical-research-council-1948/); Fisher 1935.

How rare is this? 107 patients, 19 deaths (2/3)

Streptomycin + bed rest (n = 55): 4 died

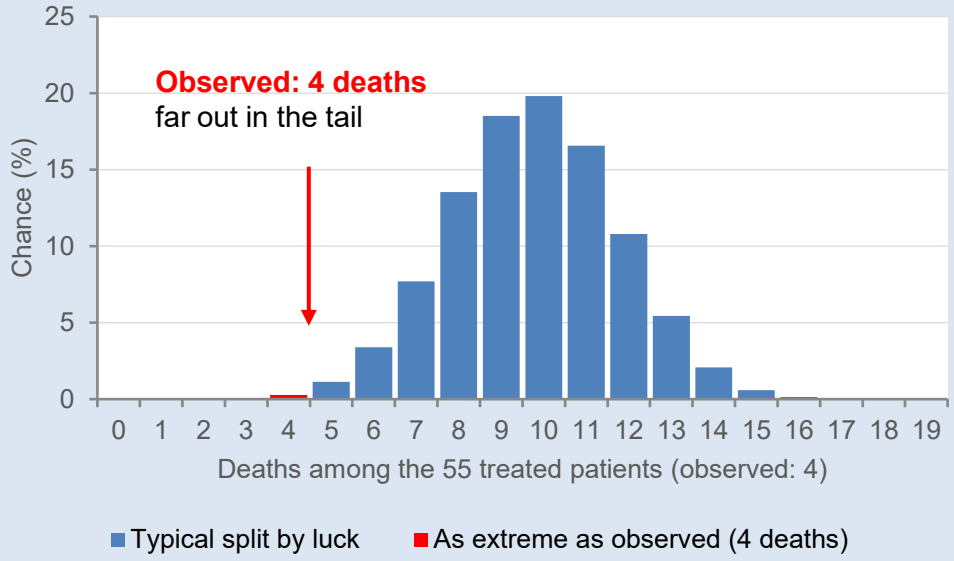


Bed rest alone (n = 52): 15 died



Red = died within 6 months (19 of 107). Each icon is one patient.

If streptomycin did nothing: deaths in its arm



5% **= 1 in 20**
≈ 5 of 107 patients

0.48% **= 1 in 200**
the observed split (Fisher exact)

References: Medical Research Council 1948; Fisher 1935 (exact test); counts via James Lind Library. Distribution and p-value: own calculation (hypergeometric).

Why biostatistics: a real trial (3/3)

MRC streptomycin trial, 1948

Design

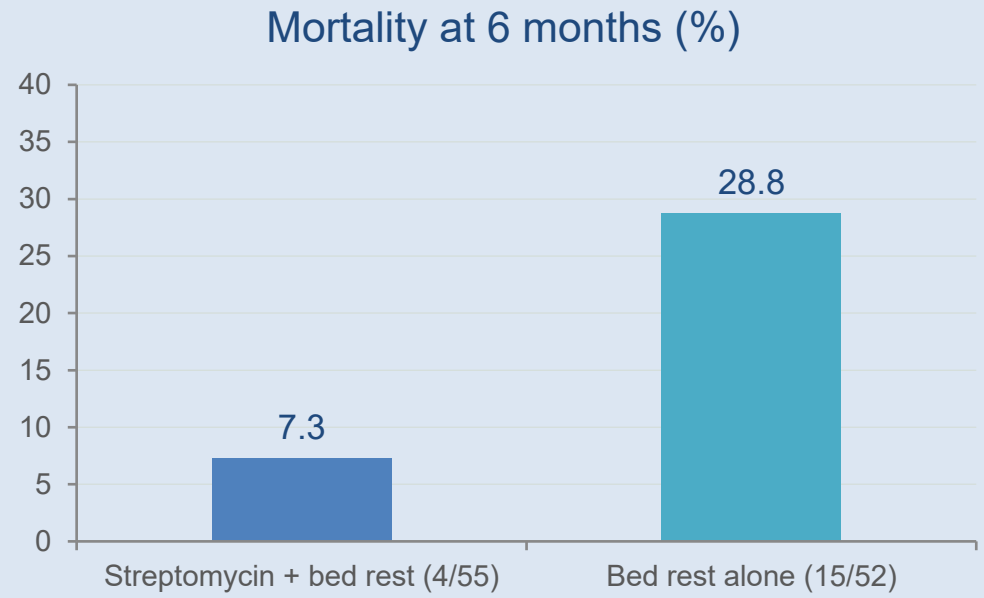
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22 pts
risk difference
95% CI 7–36

0.25
risk ratio
95% CI 0.09–0.71

$p \approx 0.005 < 0.05$
Fisher exact

Streptomycin reduces mortality!!!

References: Medical Research Council 1948 (BMJ 2:769–82); counts via James Lind Library (jameslindlibrary.org/medical-research-council-1948/); Fisher 1935.

Five eras, one continuous story

Biostatistics

early 1900s →

TAUGHT US

How to reason under uncertainty

Bioinformatics

1970s →

TAUGHT US

How to compute with biological data at scale

Machine & deep learning

1990s → 2010s

TAUGHT US

How machines recognize complex biological patterns

Generative AI

2020s

TAUGHT US

How machines interact with scientific knowledge through language

Agentic AI

emerging now

TAUGHT US

How systems take part in the scientific workflow itself

No era replaced the one before — each is layered on top of the last.

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A toolkit for separating signal from noise

Probability	Models of chance and natural variation	*Bernoulli, Bayes, Laplace, Gauss*
Estimation	Effect sizes with confidence intervals	*Neyman, confidence intervals (1937)*
Hypothesis testing	Could chance alone explain this result?	*Fisher (1925); Neyman–Pearson (1933)*
Regression	Relating outcomes to covariates	*Galton (1886); Cox model (1972)*
Experimental design	Randomization, replication, blocking	*Fisher (1935); MRC streptomycin trial (1948)*

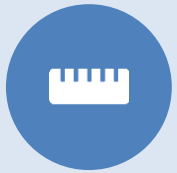
References: Galton 1886; Fisher 1925, 1935; Neyman & Pearson 1933; Neyman 1937; MRC 1948; Cox 1972.

Three disciplined questions



Is this effect real?

Bias, confounding and study design determine whether a difference reflects biology.



How uncertain are we?

Confidence intervals, variance and statistical power quantify what we do not know.



Could chance explain what we see?

Null models, p-values and multiple-testing control guard against false discoveries.

These questions never expire — they return at the end of this talk.

References: Galton 1886; Fisher 1925, 1935; Neyman & Pearson 1933; Neyman 1937; MRC 1948; Cox 1972.

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Then biology changed

DNA sequencing

Genomes, exomes and single cells, read at scale

Microarrays

Tens of thousands of transcripts on one chip (from 1995)

Mass spectrometry

Thousands of proteins and metabolites per run

Medical imaging

Millions of voxels in a single scan

Electronic health records

Millions of patients followed over decades

3.1 billion

base pairs in a single human genome

~20,000

protein-coding genes measured in one expression profile

$100M → ~$1K

approximate cost of sequencing a human genome, 2001 → mid-2010s (NHGRI)

References: Schena et al. 1995; IHGSC (Lander et al.) 2001; Wetterstrand, NHGRI sequencing-cost data.

The question itself changed

BIOSTATISTICS ASKED

“Is there a statistically significant difference?”

One comparison, one test, one answer.

BIOINFORMATICS ASKED

“How do we … enormous biological datasets?”

store | process | search | align

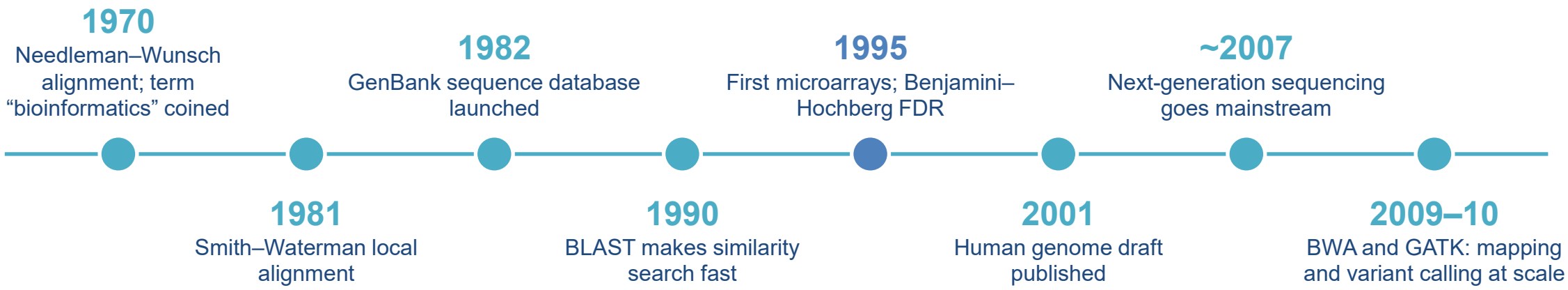
annotate | integrate | interpret

Interpretation is now the last of seven steps.

A new toolkit joined statistics

New ingredients

Algorithms | Data structures | Databases | Optimization | HPC



References: Needleman & Wunsch 1970; Hogeweg 2011; Smith & Waterman 1981; Bilofsky et al. 1986; Altschul et al. 1990; Schena et al. 1995; Benjamini & Hochberg 1995; IHGSC 2001; Bentley et al. 2008; Li & Durbin 2009; McKenna et al. 2010.

Biology became a computational science

Sequence alignment

Which sequences are related?

Genome assembly

How do we rebuild a genome from short reads?

Phylogenetics

What is the evolutionary history?

Expression analysis

What changes between conditions?

Variant calling

Which differences are real mutations?

Statistics stayed inside every pipeline: E-values in BLAST · likelihood models in phylogenetics · FDR in differential expression · genotype likelihoods in variant calling

References: Altschul et al. 1990; Felsenstein 1981; Love, Huber & Anders 2014; Benjamini & Hochberg 1995; McKenna et al. 2010.

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From specifying patterns to learning them

Instead of specifying every pattern by hand, train models to discover patterns from data.

From the late 1990s, classifiers, kernel methods and ensemble models entered genomics, proteomics and clinical research.

Classify tumors

Expression subtypes, e.g. PAM50 in breast cancer

Predict protein properties

Localization, stability and function from sequence

Prioritize variants

Pathogenicity scores such as CADD

Identify biomarkers

Sparse models (e.g. LASSO) select features

Integrate data types

Genomics, transcriptomics and clinical records

Read biomedical images

Histopathology and radiology classifiers

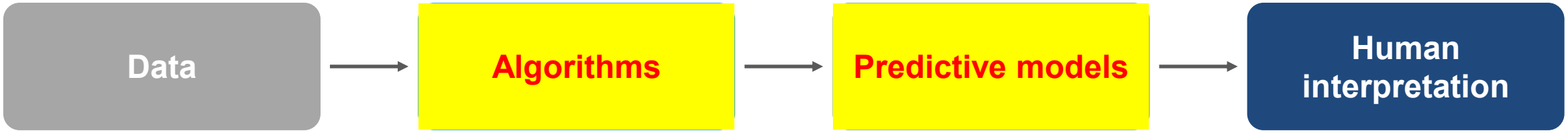
References: Golub et al. 1999; Parker et al. 2009; Kircher et al. 2014; Tibshirani 1996.

The workflow gained new stages

Classical biostatistics



Bioinformatics & machine learning



The human still chose every step — which data, which algorithm, which model, and how to read the result.

Learning representations of biology

DNA as language

Models read sequence like text and learn regulatory grammar.

EXAMPLES

DeepBind & DeepSEA (2015) · DNABERT (2021) · Enformer (2021)

Proteins as structured objects

Models learn structure and function from sequence.

EXAMPLES

ESM protein language models · AlphaFold2 (2020–21; Nobel Prize in Chemistry 2024)

Cells as high-dimensional states

Models embed single cells in learned state spaces.

EXAMPLES

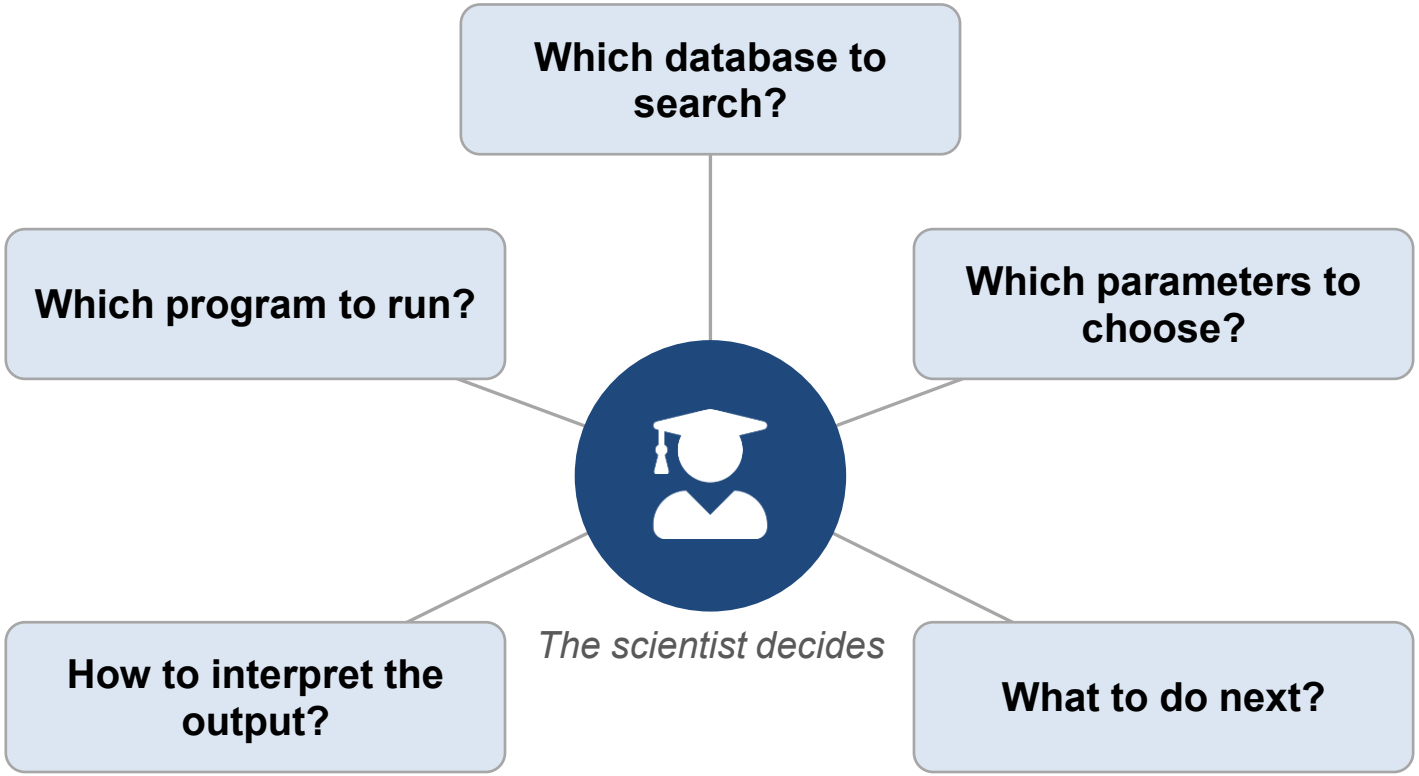
scVI (2018) · Geneformer (2023) · scGPT (2024)

References: Alipanahi et al. 2015; Zhou & Troyanskaya 2015; Ji et al. 2021; Avsec et al. 2021; Rives et al. 2021; Jumper et al. 2021; Lopez et al. 2018; Theodoris et al. 2023; Cui et al. 2024.

But AI systems were still tools

Powerful, but passive.

Every decision about the scientific workflow still came from a person. The intelligence lived inside individual steps — not in the workflow.



References: Jumper et al. 2021; Cui et al. 2024; Theodoris et al. 2023.

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Thought experiment: a new disease gene

1. Search a genomic database
2. Retrieve the gene sequence
3. Run similarity searches
4. Inspect expression data
5. Search the literature
6. Analyze pathways
7. Check known variants
8. Run statistical tests
9. Combine into a hypothesis



Each individual step might already be automated.

But the scientific workflow itself was not.

The researcher is the glue: every hand-off is a human decision.

From a question to a goal

ASK A QUESTION

“What does this gene do?”

→ **Returns an answer**

GIVE A GOAL

“Investigate whether this gene is a plausible therapeutic target for this disease.”

→ **Launches an investigation**

- **Reason**
- **Decompose**
- **Choose tools**
- **Query databases**
- **Run pipelines**
- **Inspect results**
- **Revise strategy**
- **Compare evidence**
- **Synthesize**

References: Gao et al. 2024; Qi et al. 2026.

What counts as an AI agent in biology?

DEFINITION

"An autonomy-enabled computational actor embedded within biological research workflows, capable of iterative goal-directed reasoning and decision-making."

Not agents: single-turn inference, prompt-only pipelines, one-shot RAG — no persistent state, task decomposition or feedback-driven adaptation.

Value is process-oriented, not output-oriented.

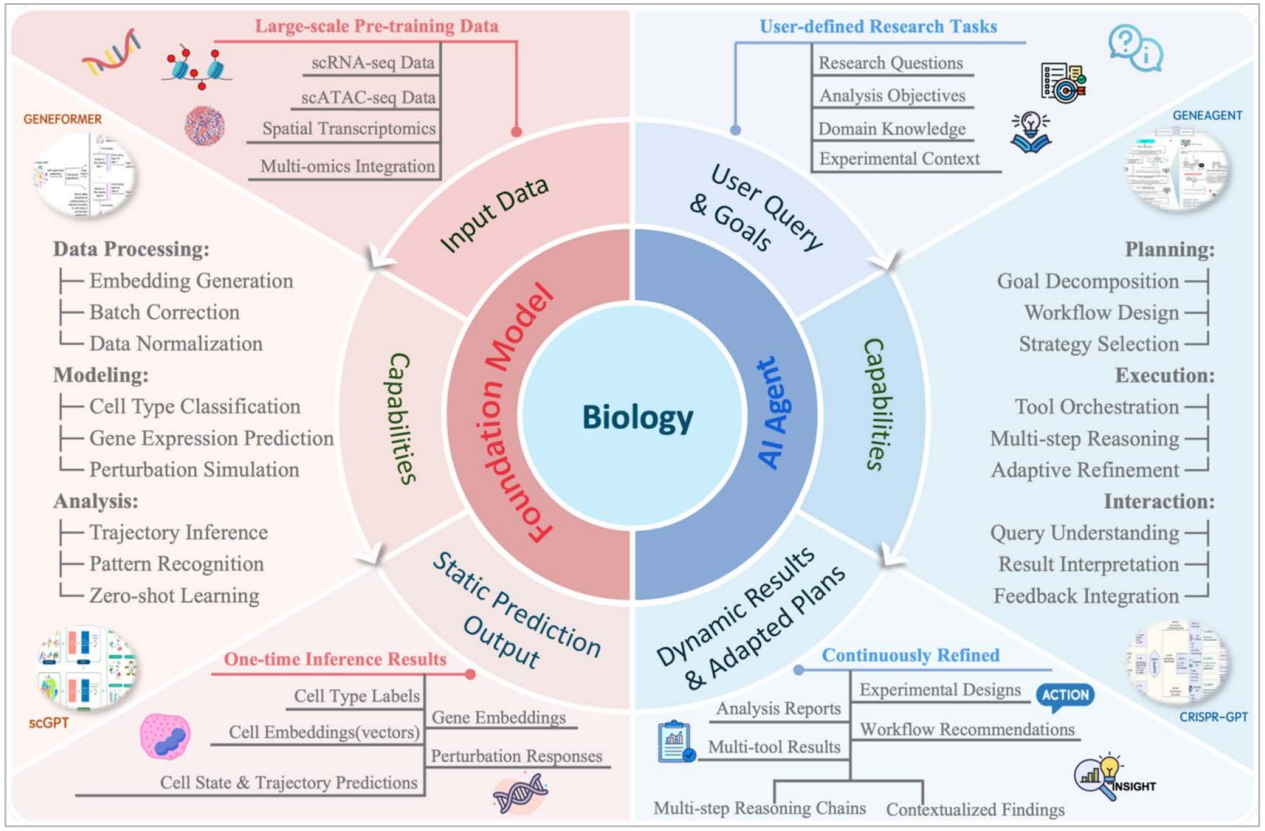
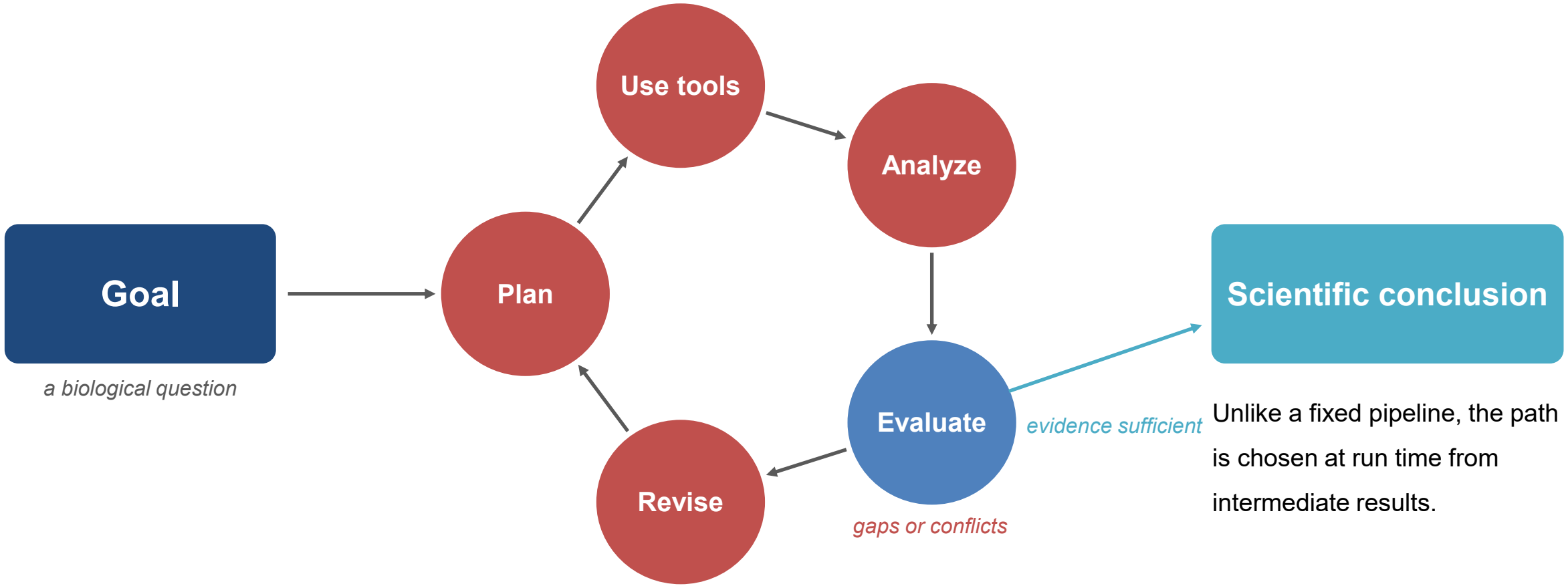


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The agentic loop

Goal → Plan → Use tools → Analyze → Evaluate → Revise → Scientific conclusion



Unlike a fixed pipeline, the path is chosen at run time from intermediate results.

References: Yao et al. 2023 (ReAct); Gao et al. 2024; Qi et al. 2026.

Anatomy of a biological AI agent

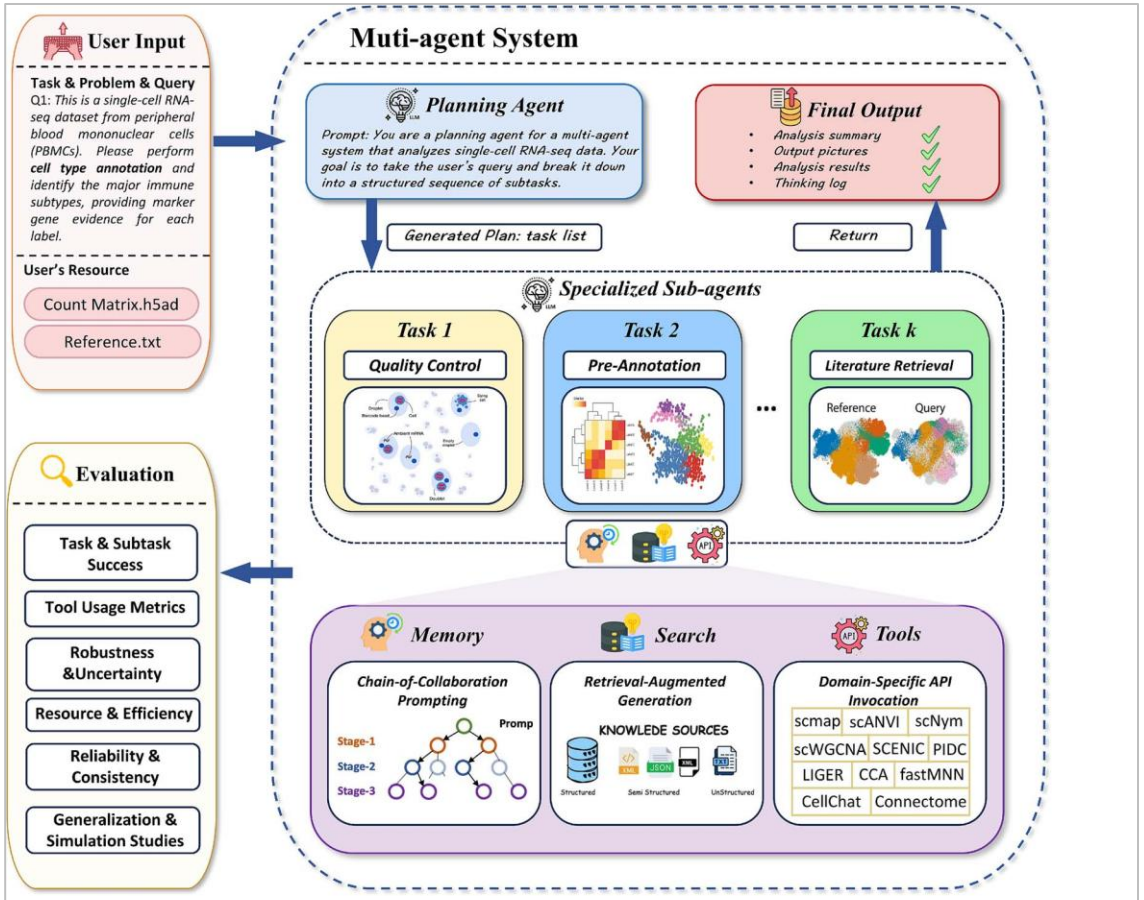


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Planning agent
Turns the user's query (e.g. annotate PBMC scRNA-seq) into a task list.

Specialized sub-agents
Quality control → pre-annotation → literature retrieval.

Memory · search · tools
Shared memory, RAG over knowledge sources, domain APIs (scANVI, SCENIC, LIGER, CellChat).

Evaluation module
Task success, robustness & uncertainty, reliability, generalization.

Closed loop
Evaluation feeds back to planning — the step that makes it an agent.

Design space: architectures and interaction

ARCHITECTURE

Single-agent

One reasoning core plans, calls tools and validates. Simple, low overhead.

GeneGPT · VarChat · MedBioLM

ARCHITECTURE

Multi-agent

Planner, executor and critic roles cross-check each other. Robust but costly to coordinate.

CellAgent · CRISPR-GPT · Biomni

ARCHITECTURE

Neuro-symbolic

Ontologies and symbolic programs constrain neural reasoning. Transparent and grounded.

AutoMAxO · Tokensome · mergen

INTERACTION

Step-by-step

Decompose → plan → execute → verify, with self-correcting loops. Interpretable.

CRISPR-GPT · ChIP-GPT

INTERACTION

Zero-shot planner

Whole plan synthesized from one structured prompt (APIs, ontology terms). Flexible.

GeneGPT · PhenoGPT

INTERACTION

Hybrid

Automation plus expert adjudication; humans resolve uncertain cases.

GPT-4 weak labels + human correction

References: Qi et al. 2026; Jin et al. 2024 (GeneGPT); Xiao et al. 2024 (CellAgent); Qu et al. 2025 (CRISPR-GPT); Huang et al. 2025 (Biomni).

Five dimensions of a biological AI agent

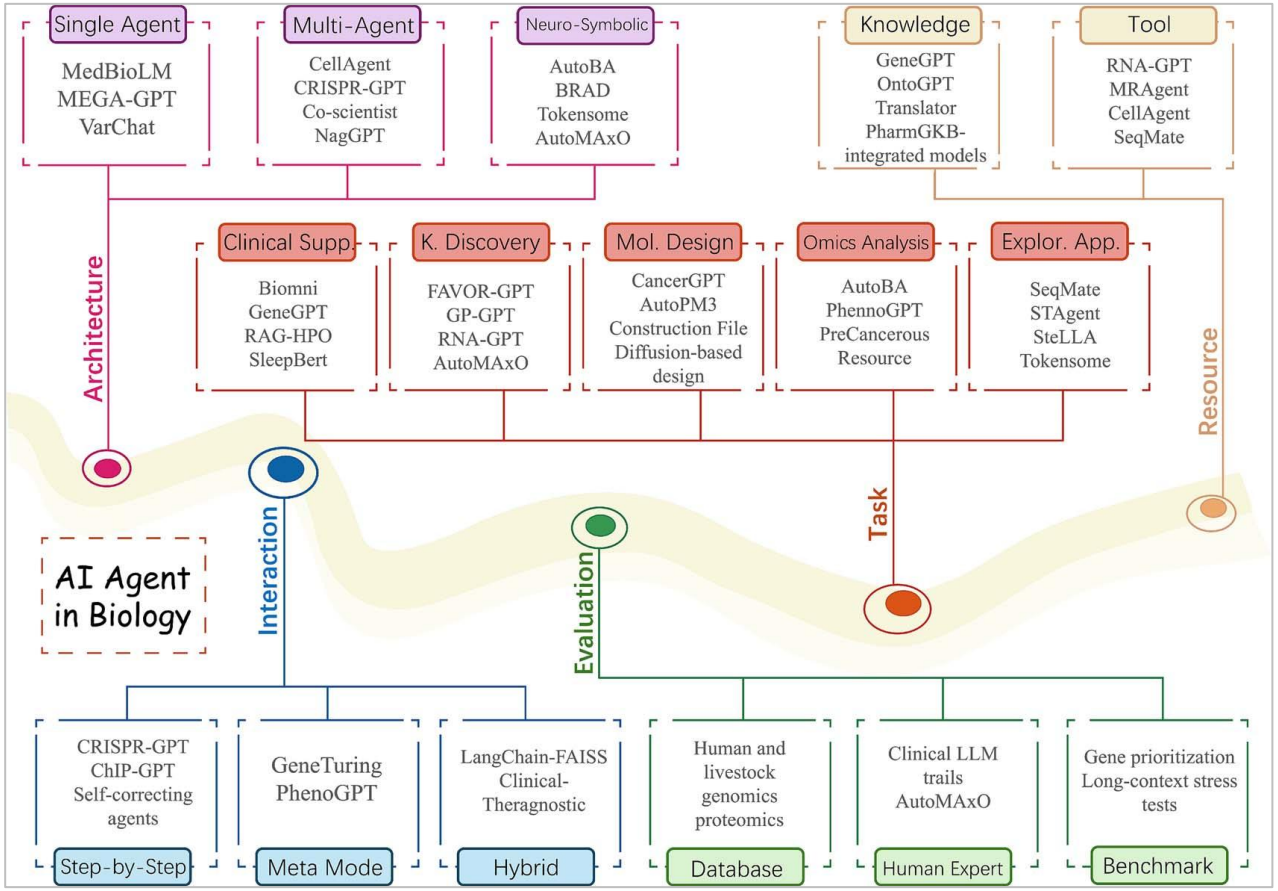
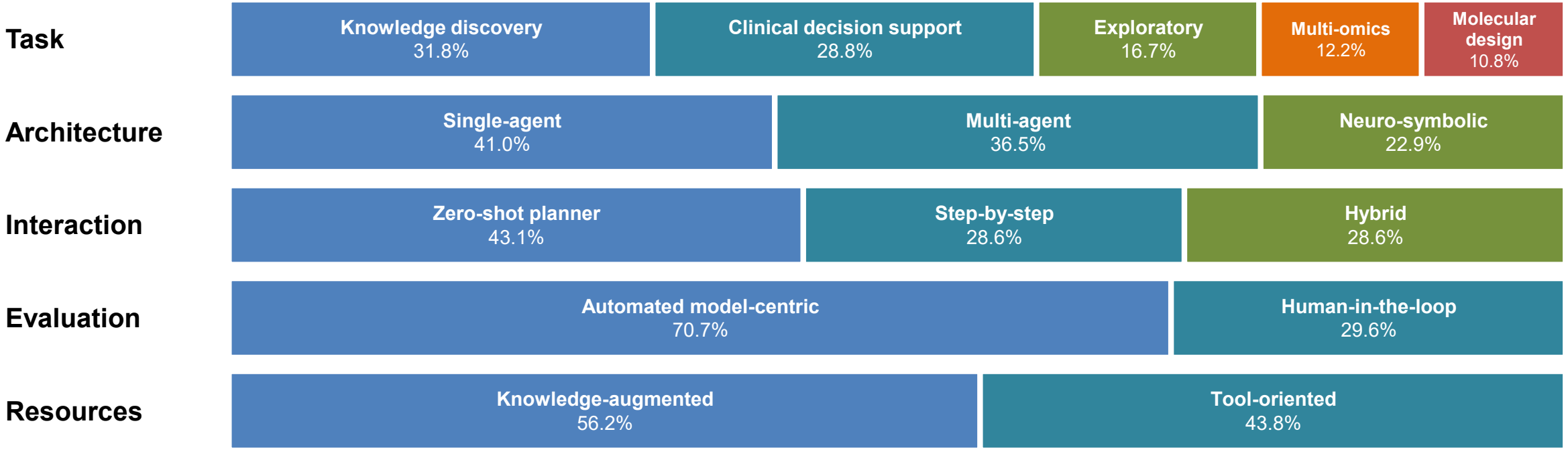


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1. **Task** *What problem?*
   Clinical · knowledge discovery · molecular design · multi-omics · exploratory
2. **Architecture** *How is cognition organized?*
   Single-agent · multi-agent · neuro-symbolic
3. **Interaction** *How does reasoning unfold?*
   Step-by-step · zero-shot planner (meta) · hybrid
4. **Evaluation** *How is success judged?*
   Automated benchmarks · human expert · benchmark suites
5. **Resources** *How are knowledge & tools used?*
   Knowledge-augmented · tool-oriented

Where the field is today



Knowledge discovery and clinical support make up ≈ **61%** of tasks; molecular design only 10.8%.

Zero-shot planning is the **most common** interaction mode (43.1%).

Only **29.6%** of systems involve humans in evaluation — most are judged by automated benchmarks alone.

Data: Qi et al., Briefings in Bioinformatics 2026 (Fig. 5); 115 studies, 2023–2025; layer-normalized shares as reported.

Representative agents at a glance

System	Task domain	Architecture		Interaction		Evaluation		Resource	
		Single	Multi	Step	Meta	Auto	Human	Knowl.	Tool
Biomni	General biology tasks	–	●	–	●	–	●	●	●
GeneGPT	Genomic QA	●	–	●	●	●	–	●	–
CellAgent	Single-cell / spatial omics	–	●	●	–	●	–	●	–
GeneAgent	Single-cell / spatial omics	–	●	●	–	●	–	●	–
CRISPR-GPT	Gene editing	●	●	●	●	●	●	●	–
BioLab	Life-science wet lab	–	●	–	●	–	●	●	●
AutoMAxO	Ontology curation	●	–	●	–	●	●	●	–
Co-Scientist	Molecular synthesis	●	–	●	–	●	–	●	●
BRAD	Biomarker discovery	●	–	●	–	●	●	●	●
AutoBA	Multi-omics automation	●	–	●	–	●	–	●	●
PhenoGPT	Pheno–geno mapping	●	–	–	●	●	–	●	–
NagGPT	Genomic query validation	●	–	●	–	●	●	●	●

READING THE MATRIX

Multi-agent designs cluster on complex workflows: single-cell, gene editing, wet lab.

Knowledge augmentation is almost universal — grounding against hallucination.

Human evaluation is still the exception.

References: Qi et al. 2026 (Table 3); Huang et al. 2025; Jin et al. 2024; Wang et al. 2025 (GeneAgent); Qu et al. 2025; Boiko et al. 2023 (Coscientist).

Multi-agent debate curbs hallucination

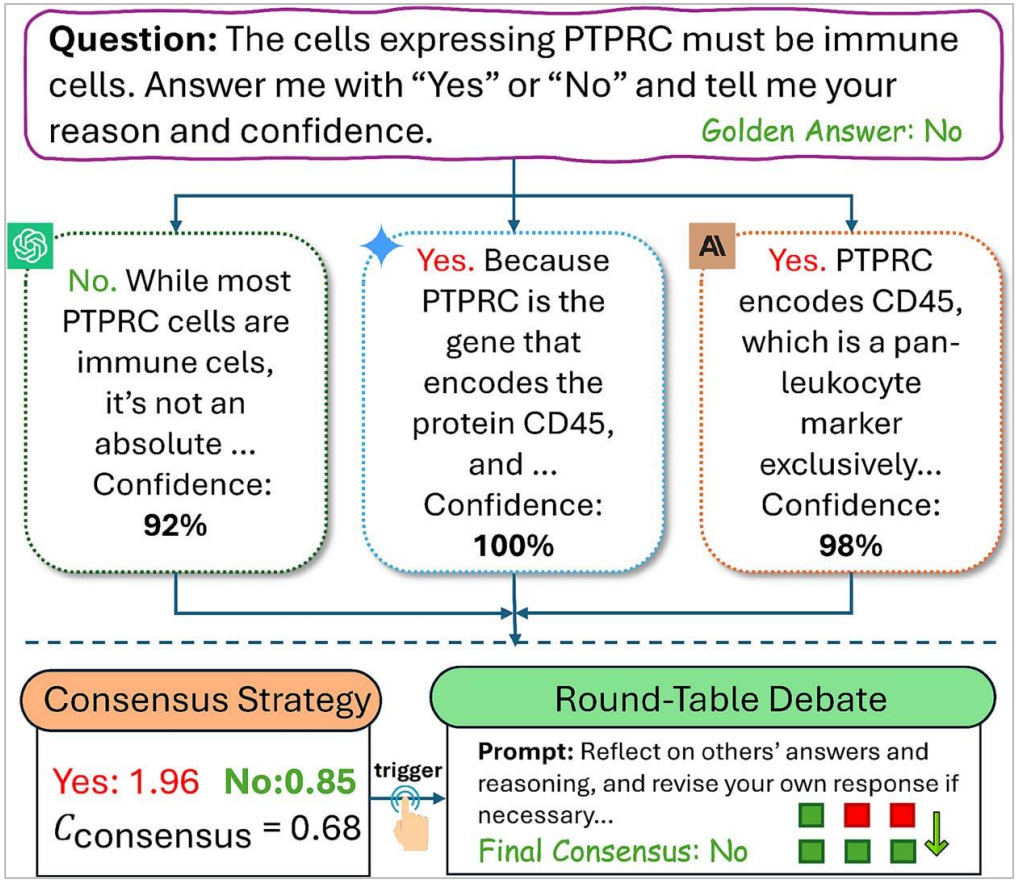


Figure reproduced from Qi C, Wang W, et al. Brief Bioinform 2026;27(1):bbag075, Fig. 4. © The Author(s) 2026, Oxford University Press; CC BY 4.0.

HOW THE DECISION WAS REACHED

1. **Majority vote**
   Yes (2 of 3 models) — wrong ✗

2. **Confidence-weighted vote**
   Yes 1.96 vs No 0.85 — still wrong: confident models dominate ✗

3. **Measure agreement, then debate**
   Low consensus (C = 0.68) triggers a round-table debate → No ✓

Lesson: disagreement is INFORMATION — quantify it, don't average it away.

An investigation, step by step

Line of evidence	Tools & data	Question the agent asks
Genetic evidence	GWAS Catalog, gnomAD, Open Targets	Is variation in the gene linked to disease risk?
Expression	GTEx, single-cell atlases	Is it expressed in the relevant tissue and cell type?
Function & pathways	UniProt, Reactome, KEGG	Does its mechanism fit the disease biology?
Druggability	AlphaFold DB, ChEMBL	Is there a pocket or existing chemistry?
Safety	DepMap, tissue expression, knockouts	Would modulating it harm healthy tissue?
Literature	PubMed	Does independent evidence agree?

References: Ochoa et al. 2023 (Open Targets); GTEx Consortium 2020; Karczewski et al. 2020 (gnomAD); Tsherniak et al. 2017 (DepMap); Varadi et al. 2022 (AlphaFold DB).



Evaluate → Revise

Finding: bulk-tissue expression is weak and inconsistent.

Re-plan: query single-cell data for the affected cell type; test GWAS–eQTL colocalization.

Output: a graded evidence report with uncertainty — not a yes/no.

Three paradigms, three questions

Traditional bioinformatics

"What computation should we perform on these data?"

Unit of work
A computation

Human role
Operator

Machine learning

"What patterns can a model learn from these data?"

Unit of work
A model

Human role
Model designer

Agentic bioinformatics

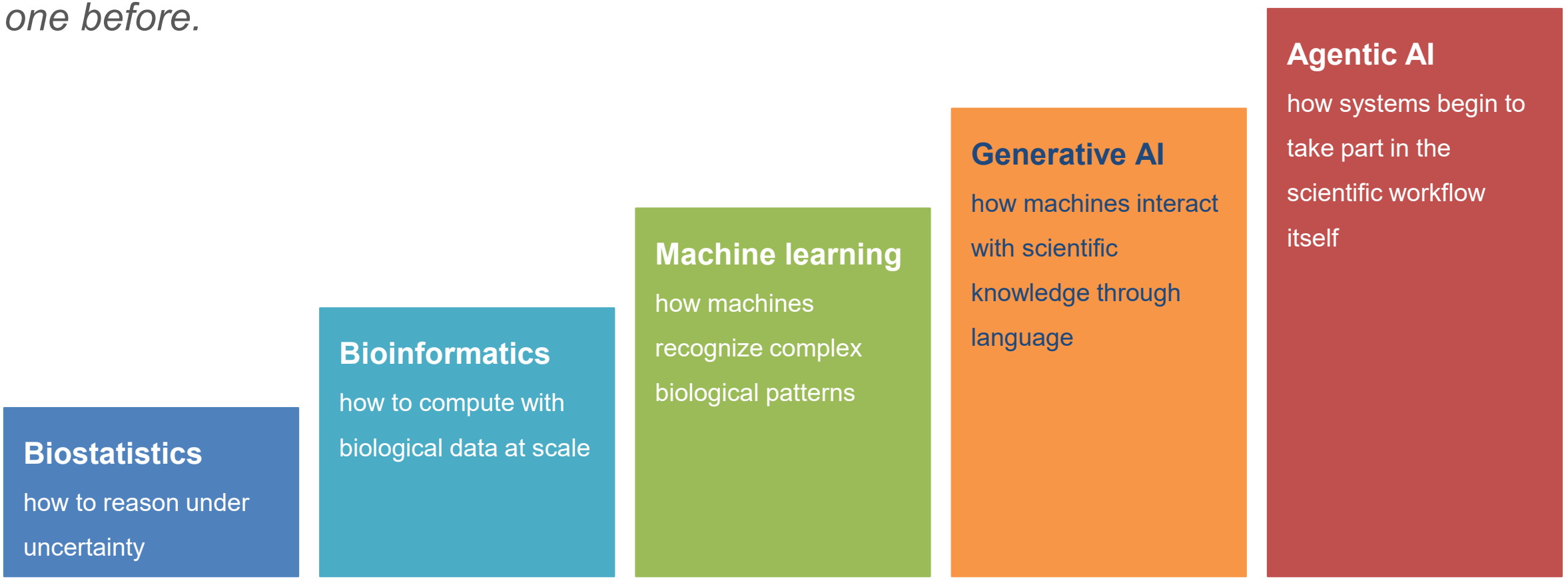
"What sequence of computational and scientific actions should be taken to answer the biological question?"

Unit of work
A workflow

Human role
Goal-setter & reviewer

What each era taught us

Each era added a capability. None replaced the one before.



Outline

- Introduction
- Biostatistics: reasoning under uncertainty
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Agentic AI in the lab: a scorecard

System	Domain	Key result	Benchmark	Expert review	Wet lab
CRISPR-GPT Qu et al. 2025	Gene editing	4-gene knockout at ~80% efficiency; activation up to 90.2%	●	●	–
Virtual Lab Swanson et al. 2025	Protein design	92 nanobodies designed; 2 gained binding to JN.1 spike	–	–	●
Robin Ghareeb et al. 2025	Drug repurposing	Ripasudil proposed for dry AMD; ABCA1 found by RNA-seq	–	–	●
AI co-scientist Gottweis et al. 2025	Hypothesis generation	AML drug hits; anti-fibrotic targets in organoids; cf-PICI mechanism	–	●	●
Biomni Huang et al. 2025	General biomedicine	Human-level on LAB-Bench DbQA (74.4%); cloning design validated	●	●	●
GeneAgent Wang et al. 2025	Gene-set analysis	1,106 gene sets; beats GPT-4; experts confirm self-verification	●	●	–
BioDiscoveryAgent Roohani et al. 2025	Perturbation design	+21% hits vs Bayesian optimization over 6 screens	●	–	–

References: Qu et al. 2025; Swanson et al. 2025; Ghareeb et al. 2025; Gottweis et al. 2025; Huang et al. 2025; Wang et al. 2025; Roohani et al. 2025.

CRISPR-GPT: gene editing on the first attempt

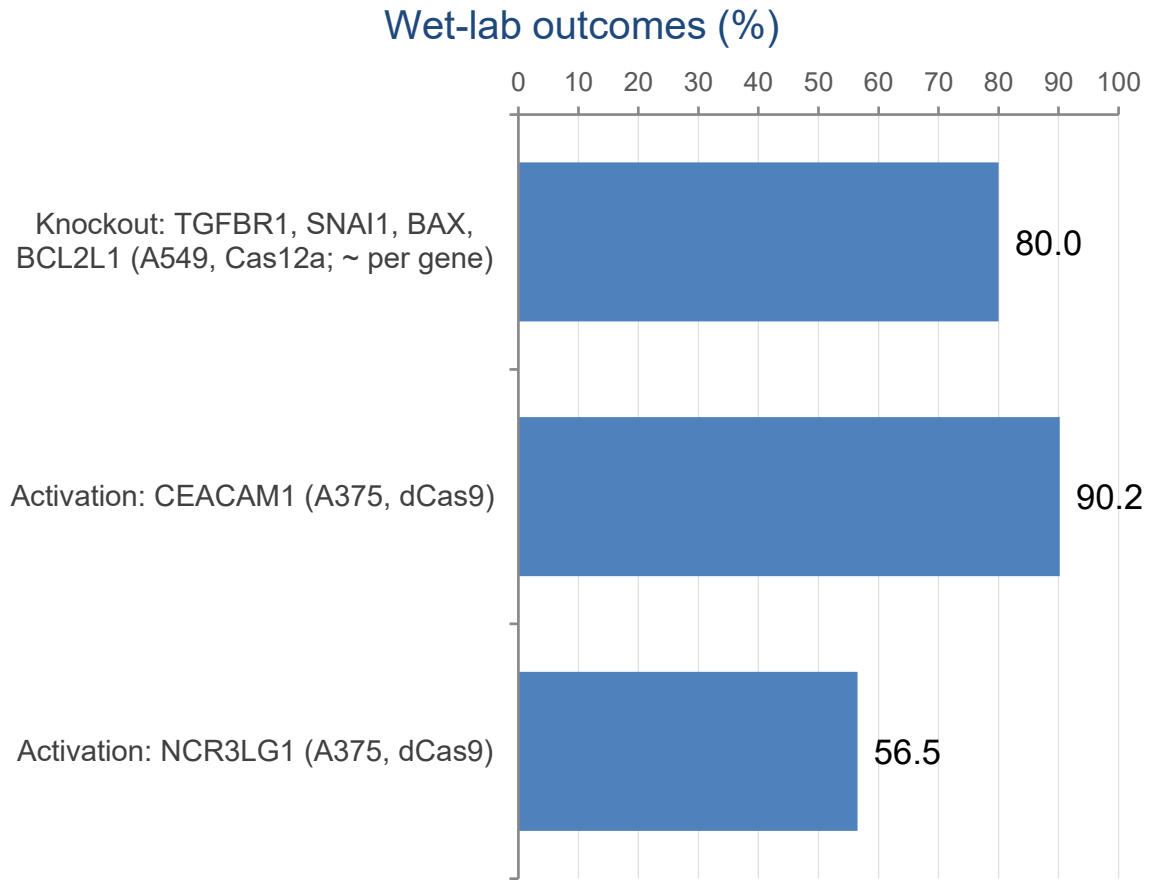
Two junior researchers, new to gene editing

— both succeeded on the first attempt.

WORKFLOW THE AGENT GUIDES

1. Choose CRISPR system
2. Design guide RNAs
3. Choose delivery
4. Draft protocol
5. Design assay
6. Analyse data (CRISPResso2)

Phenotype confirmed: after TGF-β, EMT-marker changes (CDH1, VIM) reduced up to 9× and 34× in knockout cells.



References: Qu Y, Huang K, Yin M, et al. CRISPR-GPT for agentic automation of gene-editing experiments. Nat Biomed Eng 2025;10:245–258.

Virtual Lab: an AI team designs nanobodies

AI AGENT TEAM

- **PI agent** sets the agenda
- **Immunologist** domain judgement
- **ML specialist** ESM scoring
- **Comp. biologist** AlphaFold-Multimer, Rosetta
- **Scientific critic** reviews every step

Human input ≈ 1% of all meeting dialogue

92 nanobody variants designed (4 parents)

~38% expressed at high / very-high levels

2 gained binding to the JN.1 variant spike

Tools the agents chose: ESM-1b · AlphaFold-Multimer · Rosetta.

References: Swanson K, Wu W, Bulaong NL, Pak JE, Zou J. The Virtual Lab of AI agents designs new SARS-CoV-2 nanobodies. Nature 2025.

From literature to a drug candidate

Robin
Ghareeb et al. 2025 (FutureHouse)

Reviews the literature
three agents: literature, candidate eval, data analysis

Hypothesis
enhance RPE phagocytosis in dry AMD

Candidate
ripasudil (approved ROCK inhibitor) — new for dry AMD

Validated
agent-designed RNA-seq → ABCA1; ~2.5 months end-to-end

AI co-scientist
Gottweis et al. 2025 (Google)

AML repurposing
predicted drugs cut tumour-cell viability in vitro

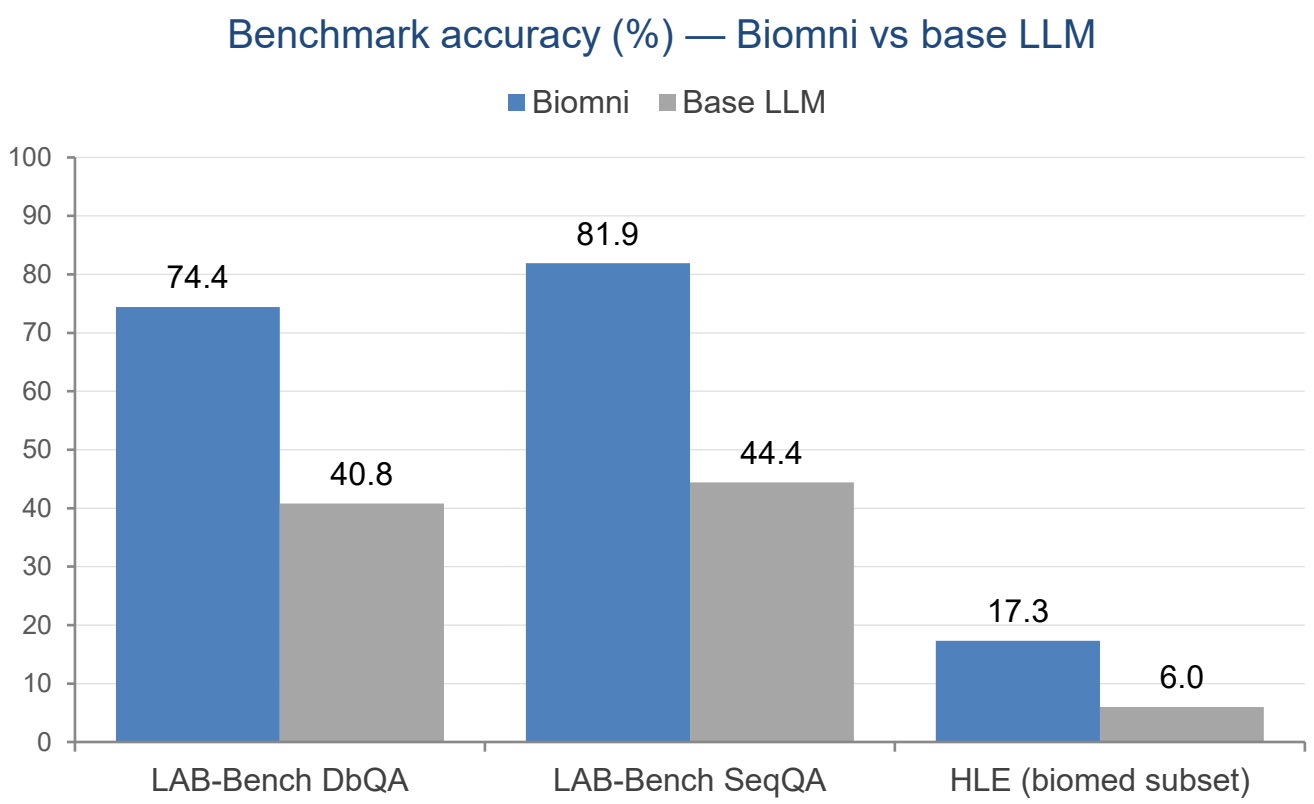
Liver fibrosis
epigenetic targets active in human organoids ($p < 0.01$)

Antimicrobial resistance
re-derived an unpublished cf-PICI gene-transfer mechanism

Speed
matched a discovery that took the lab years

References: Ghareeb AE, et al. Robin: a multi-agent system for automating scientific discovery. arXiv 2025:2505.13400. Gottweis J, et al. Towards an AI co-scientist. arXiv 2025:2502.18864.

Biomni: one agent, many tasks



Human expert on LAB-Bench DbQA ≈ 74.7% — Biomni (74.4%) essentially matches it.

ACTION SPACE

~150	specialized tools
105	software packages
59	databases
336k	single-cell nuclei analysed in a 5-hour pipeline
✓	cloning protocol validated at the bench

References: Huang K, Zhang S, Wang H, et al. Biomni: a general-purpose biomedical AI agent. bioRxiv 2025. doi:10.1101/2025.05.30.656746.

Read these results like a statistician

Mind the level of evidence Benchmark ≠ expert review ≠ wet-lab validation.

Small n 2 nanobodies, a few cell lines, one construct — promising, not proof.

Selective reporting Successes are published; failed runs usually are not.

Leakage & contamination Training on the answer inflates scores — held-out tests matter.

Replication Almost none independently reproduced yet.

THE THREE QUESTIONS RETURN

Is the effect real?

How uncertain are we?

Could chance explain it?

An agent can run 100 analyses. It still needs statistics to know which to believe.

References: Benjamini & Hochberg 1995; Ioannidis 2005; Qi et al. 2026.

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More autonomy demands more statistics

Uncertainty

Calibrated confidence, not just a confident answer

Reproducibility

Logged code, versions, parameters and seeds

Bias

Skewed cohorts, databases and literature

Validation

Held-out data, orthogonal assays, replication

Experimental design

Power, controls and confounders

Causal interpretation

Association ≠ mechanism ≠ drug target

An agent runs 100 independent tests at $\alpha = 0.05$, with no true effects…

$$\mathbf{99.4\%} = 1 - 0.95^{100}$$

chance of at least one false positive

$$\approx \mathbf{5} = 100 \times 5$$

expected false “discoveries”

References: Benjamini & Hochberg 1995; Qi et al. 2026.

Evaluating an agent is not scoring an answer

CLASSICAL BENCHMARKS

Accuracy · AUROC · sensitivity · RMSE

“Is the final answer right?”

A model can predict well and still **fail as an autonomous analyst**.

End-to-end task success	Completes the full workflow within step, time and tool budgets
Planning & re-planning	Intermediate steps serve the goal; strategy revised when needed
Tool-use reliability	Right tool, right parameters — **no violated statistical assumptions**
Evidence grounding	Claims tied to verifiable database IDs or primary literature
Error management	Catches species mismatch, gene-symbol issues, batch effects, doublets
Memory consistency	Gene lists, clusters and parameters stay coherent across stages
Long-horizon robustness	Stable under perturbation; no drift in long interactions

Source: Qi et al., Briefings in Bioinformatics 2026 (Table 2).

Open challenges and frontiers

Chat → end-to-end agents

Most systems are still query–response assistants, not goal-driven agents.

Hallucination & grounding

One wrong marker or pathway propagates through every later step.

Privacy & compliance

HIPAA/GDPR vs. powerful models that are closed and cloud-hosted.

Standardized evaluation

Case studies and small benchmarks; closed APIs block independent checks.

Cost & scalability

Long-context omics workflows exceed typical academic infrastructure.

Domain specialization

Naive vs. effector-memory T cells needs ontologies, not just text.

UNDEREXPLORED FRONTIERS

Closed-loop wet-lab experimentation

monitor, diagnose failures, redesign assays

Ecological & evolutionary systems

long horizons, scarce standard data

Temporal & regulatory processes

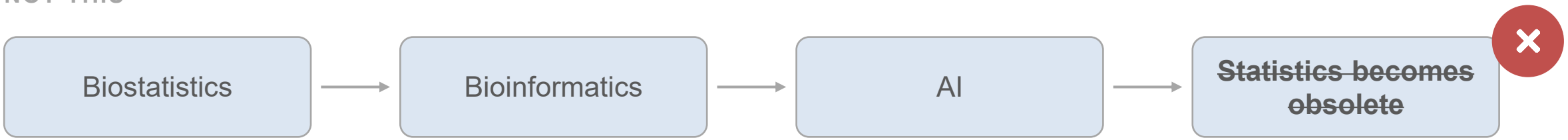
development, clinical-trial design

Directions: lightweight domain models, multi-agent robustness, open standards.

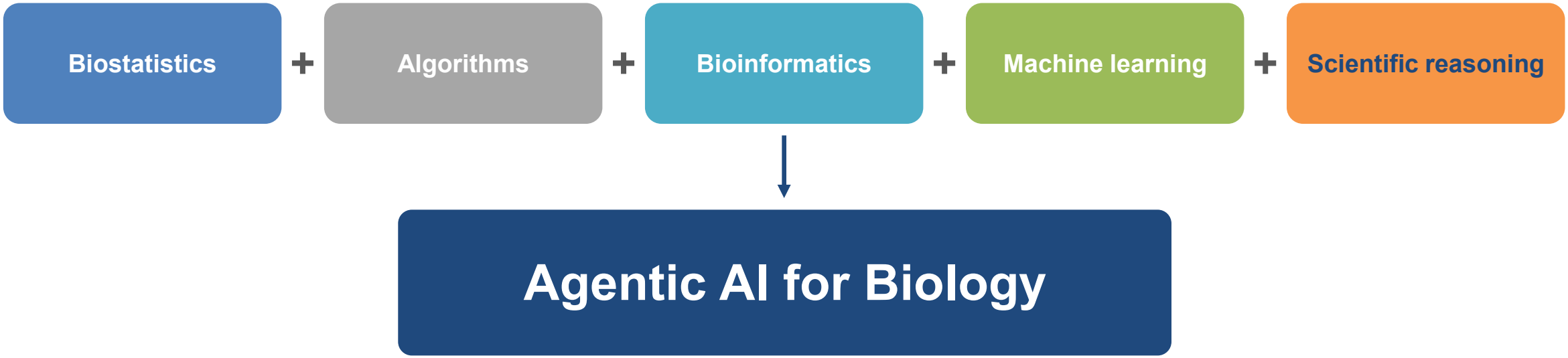
Source: Qi et al., Briefings in Bioinformatics 2026 (Discussion).

Additive, not a replacement

NOT THIS



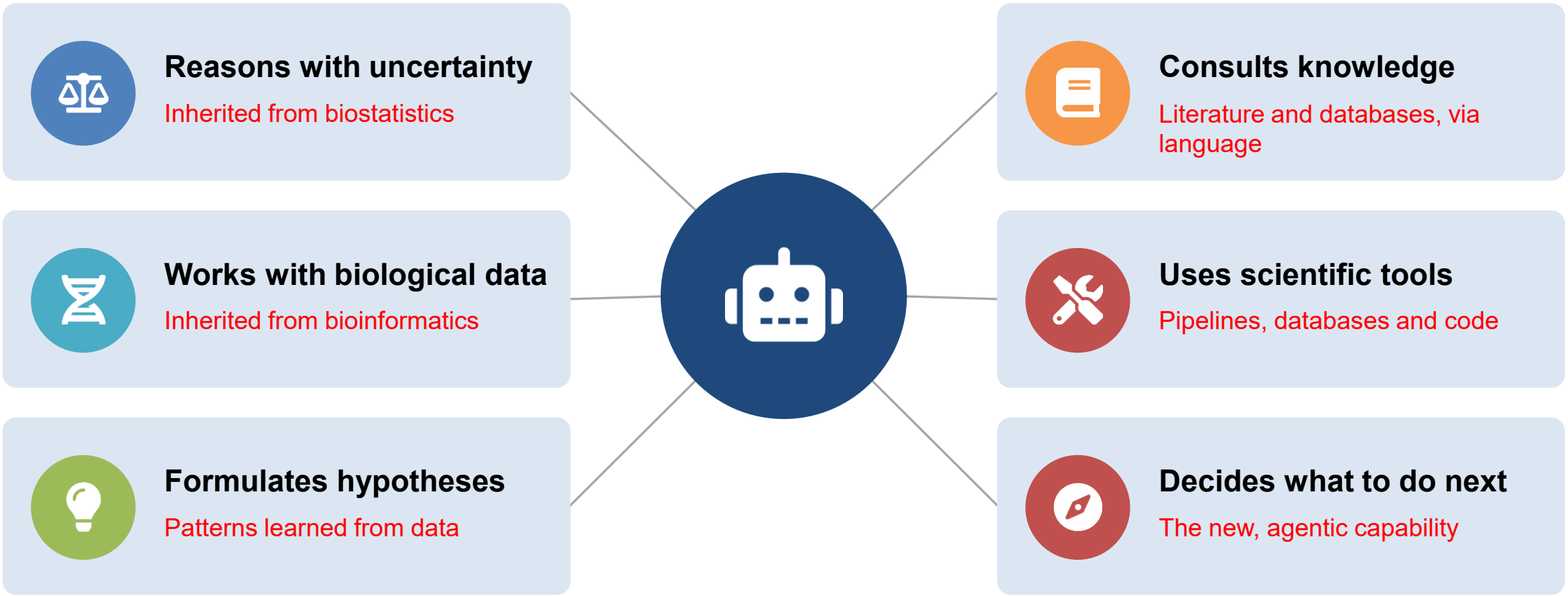
BUT THIS



Outline

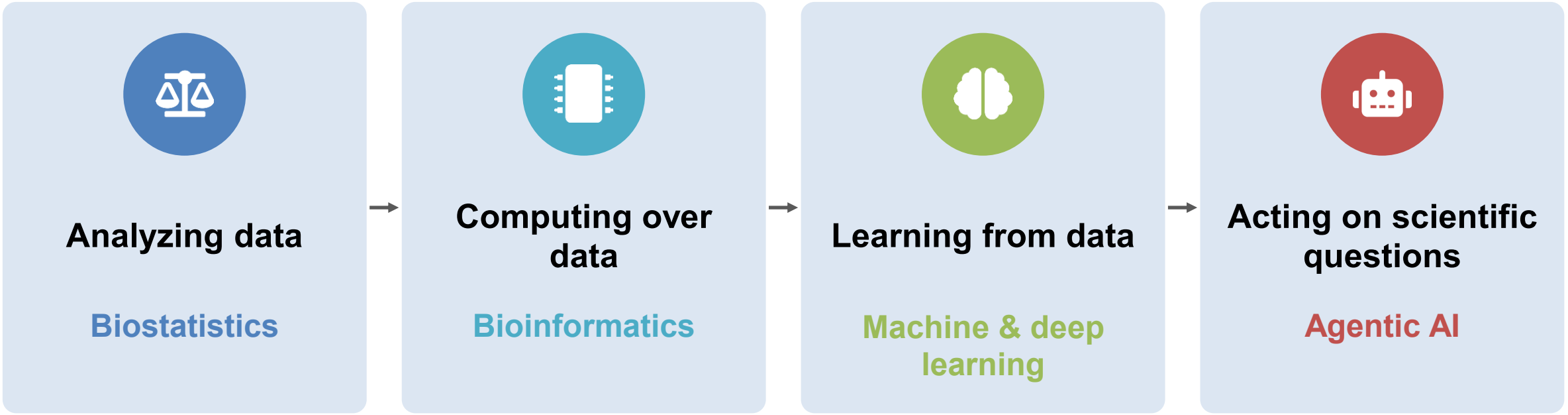
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A new kind of computational scientist



References: Gao et al. 2024; Qi et al. 2026.

From analyzing data to acting on scientific questions



The more autonomous our AI systems become, the more they need statistics, biology and human judgment — not less.

Q & A